

Physics-Informed Diffusion Models for Precipitation Nowcasting

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Abstract

Nowcasting is the forecasting of rainfall a few hours ahead from recent observations. It is the last line of defence before severe weather reaches people, yet forecasts of small, fast-moving storms are still not skilful enough. Traditional methods extrapolate the motion of rain with optical flow, but they cannot create or decay storms. Deep learning methods can learn growth and decay, but models trained with mean squared error produce blurry forecasts that lose the sharp detail forecasters need. Diffusion models can avoid this blur by generating sharp, realistic fields. This project proposes an advection-conditioned residual diffusion model for rainfall nowcasting over the United Kingdom. A semi-Lagrangian advection scheme first produces a fast physical estimate of the future rain field. A conditional diffusion model then learns only the residual, which is the growth, decay, and fine detail that advection misses. The work uses the Met Office one-kilometre radar composite. The model is compared against optical flow and a leading generative baseline, with results split by storm speed and lead time. The intended outcome is a quantified, honest account of when and by how much a prescribed physical first guess improves diffusion-based nowcasts of heavy rainfall over the United Kingdom, including the cases where it does not help.

Declaration

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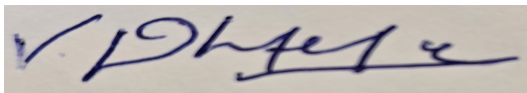
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1 Introduction

1.1 Motivation and context

Severe weather is becoming more frequent and more costly across Europe, and the short-range forecasts that protect people from it are still not good enough for the hardest cases [1]. Nowcasting the zero-to-six-hour forecast that sits directly on live observations is the last line of defence for civil protection, transport, energy, and flood-warning services [1, 2]. Its biggest weakness is the prediction of small, fast, convective storms at the sub-kilometre scale, which cause flash floods yet are predicted least well [1].

Classical nowcasting has two branches. Extrapolation of radar observations is fast and accurate in the first hour but cannot say when a storm will grow or die [2]. Numerical weather prediction can model growth and decay but is too slow to spin up in the first few hours [1, 2]. Modern nowcasting aims to combine both. The European review reports large and rising weather losses and names better nowcasts as a direct support for flood warning and public safety. It also highlights new observations, such as Meteosat Third Generation from 2023, that raise the value of methods able to turn rich data into sharp forecasts [1]. Radar remains the backbone for UK rainfall nowcasting because of its high resolution in space and time, and this project uses radar.

1.2 Problem statement and research gap

Most operational extrapolation rests on Lagrangian persistence: each parcel of rain is assumed to keep its value and only move with the wind (advection), so a motion field is estimated and the latest observation carried forward along it [2]. This works for rain that simply moves but breaks when rain grows or decays, the single largest source of error in classical nowcasting [2].

Deep learning can learn growth and decay, but networks trained with mean squared error produce blurry forecasts [2, 4]. The reason is multi-modality: a storm may follow several distinct, physically valid futures, and minimising a pixel-wise squared error drives the network towards their conditional mean, a smeared blend rather than any single realistic field [2, 4]. Generative models avoid this by sampling one sharp future. DGMR, a conditional generative adversarial network on UK radar, was preferred by expert forecasters in nearly nine of ten cases, though it has no explicit physics and decays heavy rain near ninety minutes [4]. NowcastNet added a learnable, physics-inspired evolution network (derived from the continuity equation) before generative refinement, improving extreme-rainfall skill to three hours [5]. Diffusion models are now replacing generative adversarial networks here, being more stable to train and sharp [6, 13]; residual methods such as CorrDiff [7] and DiffCast [8] make a coarse deterministic forecast and use diffusion to add the missing detail.

There are still two gaps. Initially, residual diffusion techniques use a second neural network to learn their coarse first guess [7, 8]. This work learns only the residual and makes use of an interpretable, physics-based advection scheme. Second, research that incorporates advection physics seldom reports *when* it is beneficial [5, 9]. A scale argument is presented by Metzl and co-authors: advection only becomes useful when a fast storm travels farther than the network can see in a single step over a long lead time [9]. The majority of generative studies conceal this effect by reporting a single averaged score. Both gaps are filled by this project.

1.3 Aims, objectives, and research question

The aim is to develop and evaluate a physics-informed diffusion model that produces sharp, skilful, probabilistic rainfall nowcasts over the United Kingdom, by conditioning a residual diffusion model on a semi-Lagrangian advection forecast. It is framed as one research question:

- **RQ.** Does conditioning a residual diffusion model on a physics-based advection forecast give sharper and more skilful nowcasts than (a) advection alone, (b) a diffusion model conditioned only on past

frames, and (c) a strong generative baseline such as DGMR? And does this benefit grow with storm speed and lead time, as the scale argument predicts [9]?

This is met through four objectives: **O1**, build a reproducible pipeline that prepares the radar composite into training sequences and quantify the data volume. **O2**, implement a semi-Lagrangian advection backbone with pysteps, report its own skill, and compute the residual targets. **O3**, design and train a conditional residual diffusion model and implement the baselines and ablations (pysteps extrapolation, plain diffusion, advection-conditioned full-field diffusion, and pretrained DGMR). **O4**, evaluate all models with deterministic, probabilistic, and spatial metrics, split by storm speed and lead time, with case studies of fast convective events.

1.4 Scope

The project focuses on UK rainfall from radar data only, with a horizon of up to three hours, where advection conditioning is expected to matter most [9]. The deliverable is a model design with a careful, honest evaluation, and not an operational real-time system. These limits keep the project both ambitious and achievable in the available time.

2 Background and Literature Review

2.1 Traditional nowcasting and its ceiling

Classical radar nowcasting is built almost entirely on Lagrangian persistence [2]: a motion field is estimated from a short image sequence (usually by optical flow), and the latest observation is advected forward with a semi-Lagrangian scheme [2, 3]. These tools and the STEPS method that generates a probabilistic ensemble by adding correlated noise at the smallest and least predictable scales are provided by the open-source library pysteps, which is an operational standard used by several national weather services [3]. The common ceiling is these methods move rain but can't create or decay it, so skill falls off rapidly as storms grow or fade [2].

Two further ideas from this tradition are reused here. Object-based methods track individual storm cells from frame to frame, which helps when one cell moves differently from its surroundings [2]. Stochastic schemes such as STEPS decompose the field into spatial scales and add correlated noise only at the smallest, shortest-lived scales, which are effectively unpredictable beyond a point [2, 3]. The motion field, the cell life cycle, and this small-scale noise are all things a learned model can be given or can rediscover from data [2].

2.2 Deep learning and the blurring problem

Machine learning was introduced to break this ceiling [2], but mean-squared-error training smears heavy rain into a fuzzy average of distinct futures (Section 1.2) [2, 4]. The cost is measurable: DGMR showed that blurry baselines lose effective resolution, dropping from one kilometre to sixteen or thirty-two kilometres by ninety minutes, while sharp methods keep the full texture [4]. A good method must therefore be judged on detail at every scale, not only on whether it places rain in the right location.

2.3 Generative models for nowcasting

Generative models sample a single sharp future instead of averaging. DGMR used a conditional generative adversarial network with spatial and temporal discriminators on UK radar and was ranked first by more than fifty expert forecasters in nearly nine of ten cases [4]. NowcastNet then evolved the field with a learnable, physics-inspired evolution network before generative refinement, though it learns that field end-to-end rather than prescribing it from physics [5]. Diffusion models are now the state of the art: PreDiff applied latent diffusion to nowcasting and gave sharp probabilistic forecasts [6], and Chase and co-authors independently show on satellite imagery that diffusion can generate

and decay cloud, including convective initiation [13]. On operational satellite data, the deep diffusion model of satellite (DDMS) reached a four-hour lead time for convection and added an expert-guided step that flags the cloud shapes linked to dangerous storms [10], while the cascade model SATcast combined diffusion stages with physical fields from a separate weather model to hold skill for many hours, reporting that physical fields matter more than past images as lead time grows [11]. The recurring message is that diffusion gives sharp, probabilistic forecasts and that physical information helps most at longer lead times [10, 11, 13].

2.4 Residual and physics-informed diffusion

This project is based on the residual approach. Analogously to the Reynolds mean–fluctuation decomposition [7], CorrDiff first predicts a coarse field with a deterministic network, and then a diffusion model predicts only the residual, where the smaller, lower-variance target converges faster and produces better detail. DiffCast nowcasts radar using the same split of global deterministic motion + local stochastic residual and outperforms both deterministic and probabilistic baselines on four radar datasets [8]. Both however *learn* the deterministic part [7, 8]. Instead, this project *prescribes* it from a physics-based semi-Lagrangian advection forecast [3] and learns only the residual on top, making the physical step cheap, interpretable, and directly computed rather than learned.

The scale argument leads to a testable hypothesis. A convolutional network only sees a limited patch around each point (its receptive field), where a fast storm starting outside that patch would be invisible to it, and advection solves this by moving the relevant pattern into view first [9]. The size of the effect is dependent on speed and lead time. Metzl and co-authors found only a marginal averaged gain, but when the results were split by advection speed and lead time, the benefit appeared for lead times about two hours and longer and increased with both, being largest for the fastest storms at the longest lead times (qualitatively described, not as a fixed percentage) [9]. One caveat is that their study is on satellite / lightning segmentation not radar regression. But the argument is about receptive fields and advective displacement not the sensor, so it should carry over to radar – and testing whether it does on UK radar is itself a contribution. A single averaged score can hide where a physical idea earns its place, so this evaluation is split by speed and lead time throughout.

3 Methodology

The design is shown in Figure 1: a physics-based advection step gives a deterministic first guess, and a conditional diffusion model adds the residual; the final nowcast is their sum.

3.1 Data

The project uses the Met Office UK Radar Rainfall Composite: a national map merging the radar stations across the UK and Ireland, in ODIM HDF5 format on a one-kilometre grid of 2175×1725 cells, updated every fifteen minutes, on a public Amazon S3 bucket with a two-year rolling window. Each pixel is an instantaneous rainfall rate in mm h^{-1} , where `nodata` cells are masked and `undetected` cells kept as zero. The composite is built from a network of more than fifteen radars covering over 99% of the UK and processed centrally (the Met Office “Radarnet” chain) to remove clutter and correct for attenuation, beam blockage, and beam height before reflectivity-to-rain conversion [14]. Quantitative accuracy is best within about 75 km of each radar and falls off with range as the beam rises (at 100 km it samples rain near 1.5 km altitude, so low-level rain can be missed). The network is dual-polarisation since 2018 and quality-controlled, though residual artefacts remain (Section 6.3) [14]. Initial work has confirmed the ingest pipeline (streaming, metadata, masking, plotting with pysteps).

Sequences use four input frames at fifteen-minute spacing (the previous hour) and up to twelve target frames (three hours ahead). Training uses 256×256 km patches rather than the full grid for memory reasons, with importance sampling towards heavy rain (as in DGMR [4]) and a time-based train/validation/test split so no future information leaks in. At a fifteen-minute cadence the archive

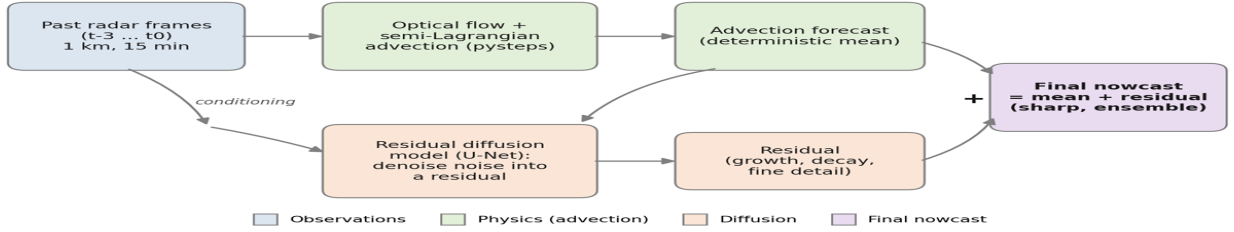


Figure 1: The advection-conditioned residual diffusion pipeline. pysteps optical flow and semi-Lagrangian advection produce the deterministic forecast A at each lead time. A conditional U-Net diffusion model, conditioned on the past frames and A (dashed arrow), denoises random noise into a residual r , giving the final nowcast $\hat{y} = A + r$. Several independent noise samples give an ensemble.

holds roughly 70,000 frames over two years (about 90 GB compressed), from which order 10^4 – 10^5 rainy patch sequences are drawn. Exact counts will be fixed by a listing pass in WP1 and reported, since they bound feasibility and the rarity of the extremes against which the model is hardest to test.

3.2 The advection backbone

The backbone is built with pysteps [3]: optical flow estimates the motion field from the four input frames, and a semi-Lagrangian scheme extrapolates the latest frame to every lead time. This is the operational Lagrangian persistence [2, 3], the step NowcastNet learns internally [5] and that Metzl and co-authors feed in as an input [9]. The motion-field magnitude gives the per-case storm speed used to split the evaluation. Because the method rests on this first guess, the advection forecast’s own skill is reported with the same metrics as a floor that any learned model must beat, and weak or inconsistent motion fields (where Lucas–Kanade struggles) are flagged, which is exactly the convective-initiation cases the residual must handle.

3.3 The residual diffusion model

Similar to CorrDiff [7] and DiffCast [8], each future field is separated into the advection forecast and residual (growth, decay, fine texture). The residual is generated by a conditional diffusion model conditioned on the past frames and the advection forecast at the target lead time, so the model always knows where the rain is forecast to move before adding detail. The Reynolds analogy directs the design, but it is not an exact correspondence because the advection forecast is a deterministic physical estimate rather than a statistical mean and the residual is systematic growth/decay rather than zero-mean noise. Instead of learning the entire field, it purchases a smaller, lower-variance target [7, 8]. Rainfall is heavy-tailed and mostly zero, so we pass it through a logarithmic transform before the split, compressing the dynamic range and keeping the residual small [7]. During generation, the model denoises a random noise to a residual, and adds it to the advection forecast. Multiple draws generate an ensemble [4, 6]. The model first jointly predicts all lead times. The two-stage cascade is the fallback if the errors from the far frames contaminate the near ones, similar to SATcast [11].

3.4 Training

Training uses the standard denoising objective (predicting the added noise) [12] with a U-Net backbone [6, 8]. Because full ensemble sampling each epoch is too costly, the cheap denoising loss is monitored every epoch for divergence, and every few epochs a small fixed-seed validation ensemble (for example eight members on a fixed subset) is scored by CRPS and CSI at a low threshold and the best-CRPS checkpoint is kept, with early stopping after a fixed patience. Fixing the validation seeds removes sampling randomness from model selection. Initial hyperparameters follow published work and are

tuned on validation only: four input frames and twelve lead times on 256×256 km patches; a U-Net trained with the ε -prediction objective over 1000 diffusion steps and 50–250 DDIM sampling steps; AdamW at learning rate 10^{-4} ; conditioning dropout 0.1 for classifier-free guidance; and a log transform of rainfall to keep the residual small [4, 8, 6, 12].

3.5 Baselines and ablations

Three external baselines and two internal ablations separate the two roles the advection forecast plays, a *conditioning input* and the *additive base* of the residual. The baselines are Lagrangian persistence / the pysteps STEPS ensemble [3] (the physical floor) and pretrained DGMR [4] (the strongest UK-radar generative model, designed for 0–90 min, so longer leads are reported descriptively). The ablations are: **M0**, plain diffusion on past frames only (isolates whether conditioning helps at all), **M1**, advection-conditioned diffusion predicting the full field (isolates advection as a conditioning input), and **M2**, the proposed advection-conditioned residual model (M2 vs M1 isolates the residual formulation). A test-time ablation also replaces the advection condition with a blank or degraded (persistence) field to measure how much the model leans on it. Together these comparisons separate whether conditioning helps from whether the residual formulation helps, which a single plain-diffusion baseline cannot do on its own.

3.6 Implementation and reproducibility

Code is in Python and released openly, building on pysteps [3] and the open DiffCast code [8] so the residual machinery need not be written from scratch. Every experiment is tracked and reproducible from a fixed data snapshot and random seed; the seed and number of samples are recorded for any reported score, since diffusion output is random by design.

4 Evaluation Plan

4.1 Metrics

Three metric families are used, following DGMR [4]. Deterministic skill uses the Critical Success Index, the probability of detection, and the false alarm ratio at thresholds such as 1, 4, 8, and 16 mm h^{-1} and several spatial pooling scales; the Critical Success Index is chosen because it balances hits, misses, and false alarms at a rain threshold. Probabilistic skill uses the Continuous Ranked Probability Score, the probabilistic generalisation of mean absolute error, which rewards a sharp *and* well-calibrated ensemble, together with rank histograms and reliability diagrams that check whether the spread is honest rather than over-confident. Spatial realism uses the radially averaged power spectral density, which exposes whether a method preserves small-scale texture or smooths it away [4]. All three families matter, because a model can score well on one and badly on another.

4.2 Storm speed and lead time analysis

This is the central, most original part of the evaluation. Every forecast is sorted by storm speed (from the motion field) and lead time and scored within each group, a direct test of the scale argument [9]. Cases are placed into slow (below about 10 ms^{-1}), medium (about $10\text{--}25 \text{ ms}^{-1}$), and fast (above about 25 ms^{-1}) bands following the scale study [9], and the twelve lead-time steps give a speed-by-lead-time grid, presented as a heat map of the skill gain of the proposed model over the plain-diffusion baseline. Significance is established by bootstrapping over test cases (resampling with replacement and recomputing the score). A gain is reported only if its confidence interval clears zero, which matters most in the sparse fast band, where intervals are widest. Alongside the aggregate grid, individual case studies of fast convective events are inspected, since they show directly whether the residual adds the growth and decay that advection misses, and where it does not.

The strategy does not assume success. A null or negative result has a predetermined meaning: the central hypothesis for UK radar is not supported if no gain clears zero in the fast, long-lead bin, and this is reported as a finding rather than concealed. A portion of the research question would be addressed by the examination of potential explanations, such as UK storm speeds rarely reaching the fast band, an already sufficient receptive field at this resolution, or advection errors dominating the residual.

4.3 Success criteria

The project succeeds if (i) the proposed model beats Lagrangian persistence and plain diffusion on CSI and CRPS at lead times beyond about 60–90 min. (ii) it keeps a power spectrum close to the real radar, showing it does not blur. (iii) the advection benefit grows with storm speed and lead time, with at least one speed-by-lead-time bin showing a gain whose bootstrap interval clears zero. And (iv) case studies show the residual adding the growth and decay that advection alone misses, and identify where it does not. Matching DGMR within its ninety-minute horizon would be a strong result, because DGMR was not built to run to three hours, the long-lead comparison is reported descriptively rather than as a claim of superiority.

5 Contribution

The project makes three contributions. First, to our knowledge it is the first to use a prescribed, non-learned physics-based semi-Lagrangian advection scheme as the deterministic base within a residual diffusion framework for radar nowcasting [7, 8], distinct from NowcastNet, whose evolution network is physics-inspired but still learned end-to-end [5]. Because the physics step and the diffusion step are separate, the first guess is swappable, where the same residual machinery could later take a numerical weather prediction field at longer ranges or a satellite-derived motion field where radar is unavailable, the direction DDMS and SATcast point towards [10, 11]. Second, it is the first test of this residual diffusion idea on the Met Office one-kilometre UK composite, the data on which DGMR was proven [4], enabling a direct and fair comparison. Third, it makes the scale argument [9] measurable for generative nowcasting on radar, which most studies do not do and which has not previously been tested in this modality.

The interpretability benefit is concrete rather than rhetorical. Because the base A is a known advection field, the residual $r = \hat{y} - A$ can be mapped directly to where and when the model adds rain (growth and initiation) or removes it (decay), correlated with the storm speed and with simple convective indicators, and attributed to specific physical regimes. With a learned deterministic base this split is not physically labelled, so the same inspection is not available. The work focuses on the precise weakness identified in the European review: skillful, sharp, probabilistic nowcasts of fast convective rain [1], and is made feasible by the light advection backbone.

6 Project Management

6.1 Work breakdown and timeline

The project runs 30 May–13 August 2026 (presentation noon 7 August, and final report noon 14 August), in six packages over roughly eleven weeks, with later packages overlapping earlier ones (Table 1). The minimum viable result is WP1–WP3 (a working model and first comparison); extra lead times and metrics are stretch goals. WP5 finishes about 5 August so results are ready for the presentation.

6.2 Required resources

Data is free and public, and access is confirmed. Compute is the main cost and is provided by the JASMIN Orchid GPU cluster (a ~ 14 -node SLURM cluster of NVIDIA A100-SXM4 40 GB GPUs),

Table 1: Work breakdown and timeline (2026).

Work package	Tasks	Dates	Deliverable
WP1: Data	Stream/prepare the composite; masking; sequence building; time split; data-volume audit	30 May – 12 Jun	Clean dataset and loader; data-volume report
WP2: Backbone	Optical flow and semi-Lagrangian advection (pysteps); advection-skill baseline; residual targets	8 Jun–16 Jun	Advection baseline and residual targets
WP3: Diffusion model	Build and train the conditional residual diffusion model; first ensembles	15 Jun – 10 Jul	Working model, first samples
WP4: Tuning + ablations	Tune on validation; latent-diffusion go/no-go; M0/M1/M2 ablations	7 Jul – 31 Jul	Tuned model and ablation runs
WP5: Evaluation	Metrics; speed-by-lead-time grid; bootstrap intervals; case studies	27 Jul – 5 Aug	Results and figures
WP6: Writing + dissemination	Write throughout; prepare the presentation	13 Jul – 13 Aug	Presentation (7 Aug); report (14 Aug)

accessed via the `orchid` partition, account, and QoS; the default QoS caps a job at 24 h wall-time (48 h on request via `orchid48`), one A100 per job. With assumptions stated for checking: one full training run (~ 200 k steps) is about 12–24 h on one A100, so ~ 20 –40 GPU-h per run with restarts; tuning and the M0/M1/M2 ablations (~ 8 –12 runs) add ~ 300 –500 GPU-h; evaluation sampling ($\sim 1,000$ cases, ~ 20 members, twelve lead times, DDIM) adds ~ 30 –60 GPU-h — a total planning budget of roughly 400–600 GPU-h, realistic for an MSc allocation. The binding constraint is the 24 h wall-time, not total hours, which is why frequent checkpointing and the latent-diffusion fall-back are built in. The two-year snapshot (~ 90 GB compressed) sits in a JASMIN group workspace; software is open source (Python, pysteps [3], a deep-learning framework, and the DGMR and DiffCast code [4, 8]).

6.3 Risk register

Table 2: Risk register (principal risks).

Risk	Likeli.	Impact	Mitigation
GPU access / jobs exceeding 24 h wall-time	Medium	High	Checkpoint/resume to fit <code>orchid</code> windows; request <code>orchid48</code> ; use latent diffusion [6], smaller patches, fewer sampling steps; start small.
Diffusion training unstable / hyperparameter-sensitive	Medium	High	Start from published DiffCast/PreDiff/DDPM settings [6, 8, 12]; small sanity runs first; change one setting at a time; plain diffusion as control.
Radar data-quality artefacts (beam blockage, attenuation, clutter, bright band, Z–R uncertainty)	Medium	Medium	Use the Radarnet quality-controlled product [14]; mask <code>nodata</code> , keep <code>undetected</code> as zero; restrict to reliably covered regions; track residuals, since the advection step inherits them.
Optical-flow failure for rapid/complex motion	Medium	Medium	Report advection skill as a floor; flag low-confidence motion fields; these are the convective-initiation cases of interest.
Extreme-rain tail rare even after importance sampling, and short history	Medium	Medium	Importance sampling plus threshold-stratified metrics; bootstrap intervals for sparse bins; store a fixed snapshot early; honest tail reporting.
Ensemble under-dispersive, as in CorrDiff [7]	Medium	Medium	Rank histograms and reliability diagrams; tune sampling noise; explore output rescaling.
Time overrun	Medium	High	Keep WP1–WP3 as the minimum result; treat extra lead times/metrics as stretch goals.

6.4 Ethical, legal, and regulatory aspects

The project uses only public meteorological data, with no human participants and no personal data, so ethical risk is low and no ethics approval is expected, and the radar data is used within its licence terms [14]. The main concern is the energy cost of training and sampling diffusion models, which is known to be high [7]. This is managed by small experiments, the latent option [6], reusing pretrained DGMR rather than retraining it, and honest compute reporting. Forecast uncertainty will be communicated clearly and not overstated, in line with the user needs raised in the European review [1].

7 Discussion

This proposal’s strength is that it provides a cheap, comprehensible physical first guess in place of a second network, tested on the same data and against the same expert-grade baseline as DGMR [4]. This is the strongest current idea, the residual diffusion model [7, 8]. There are still a few restrictions. The residual approach makes the assumption that advection is a good first guess; for convective initiation, the motion field carries little signal, and the residual must do nearly all the work. This is the hardest case, which the evaluation reports the advection forecast’s own skill (Section 3.2). The latent route is a planned decision with a concrete trigger at the end of WP3 (switch to a PreDiff-style latent model with a pretrained autoencoder that compresses the field before diffusion [6] if pixel-space training will not fit one A100 or one run cannot be made to fit 24h even with checkpointing). Diffusion sampling is slow, so the method is not real-time. Since DGMR revealed that standard scores and expert usefulness do not always agree [4], the two-year window restricts the extremes available, and verification is itself open, multiple metrics are used, but a comprehensive expert study is outside the scope. Although the design is intended to extend to satellite or NWP first guesses, results may not directly transfer because the work is limited to the UK and radar [10, 11, 13].

There is also a real tension between sharpness and calibration. A diffusion model gives an ensemble, not a single answer, and the value of that ensemble depends on whether its spread is honest, an over-confident ensemble, as CorrDiff’s was found to be [7], would mislead a forecaster who trusts the spread, which is why rank histograms and reliability diagrams are treated as first-class results here and not as extras, and why the plan includes tuning the sampling noise or rescaling the output if the spread is too narrow. The central hypothesis may also simply not hold on UK radar. If the speed-by-lead-time split shows no significant gain, that is a genuine, reportable outcome about the physical scales of UK convection relative to the model’s field of view, not a failure of the project. Showing that the residual idea works with a simple, transparent first guess tests the core claim in its cleanest setting before more complex first guesses, such as a numerical weather prediction field or a satellite motion field, are added [10, 11].

8 Conclusion

This proposal sets out a clear, feasible project: it gives the residual diffusion model [7, 8] a prescribed, physics-based first guess from a semi-Lagrangian advection scheme [3] rather than a second network, applies it to the Met Office UK radar composite on which DGMR was proven [4], and evaluates it with a storm-speed and lead-time analysis that makes the scale argument [9] measurable on radar. If it succeeds, it shows a cheaper, more interpretable route to the sharp, skilful, probabilistic nowcasts of fast convective rain that matter most for public safety [1]; if it does not, the same evaluation will say so clearly. The plan, resources, risks, and ethics are all manageable within the time available.

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